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HASTHAS™ SCIENTIFIC INSTRUMENTS INDIA PVT. LTD.,

PRODUCT CATALOGUE

Technical Specifications

BOD Incubator

Muffle Furnace

Laminar Flow - Horizontal

Deep Freezer-Horizontal

Manufacturer of Laboratory and Scientific Instruments

Compiled from the supplied product brochure and technical specification document

About This Combined Catalogue

This catalogue combines the product photographs from the supplied brochure with the corresponding technical specifications from the supplied specification document. Products were matched by their printed names and model descriptions. Where one specification covers multiple pictured variants, the variants are grouped together. The General Terms and Conditions from the supplied file are retained at the end of this catalogue.

Brochure products reviewed	**40**
Main matched catalogue entries	**28**
Grouped pictured products	**33**
Products requiring separate specifications	**7**

Important: Entries marked as a common or closest match should be checked by the manufacturer before commercial printing or quotation use.

01. Autoclave - Vertical Models

With controller and auto-pressure control

Without control system

Specification reference: HSI-101

The supplied specification file contains one common HSI-101 vertical autoclave specification. Controller and automatic pressure-control features are listed under optional accessories.

Technical Specifications

HSI-101 AUTOCLAVE VERTICAL

Vertical Autoclave, high pressure electrically heated steam sterilizer is used for sterilization of hospital surgical instruments, dressing material, linen, rubber, plastic materials by mean of saturated steam under pressure of 20 psi. HASTHAS Autoclave is also used in all scientific laboratories, educational institutions and industries.

Construction: Double walled construction with boiler made of stainless steel thick sheet. Inner and Outer chamber is fully made of Stainless Steel Sheet. Lid is made of stainless steel thick plate and is tightened all-round by wing nut locking system. Sterilizer is hydraulically tested upto 40 psi as safety measures and fitted with Neoprene Rubber Gaskets to ensure tight sealing.

Autoclaves are fitted with pressure gauge, steam release cock, spring-loaded safety valve that can set at any selected point from 15 psi to 20 psi +/- 3 psi and drain outlet. An ISI marked immersion type suitable rating heating element heats the water and steam to desired temperature and pressure.

Supplied complete unit with PEDAL LIFTING DEVICE, Perforated S.S. Basket, Wing Nut opener, Cord and plug to work on 220 Volts, 50 Hz AC supply

Maximum Pressure : 30 psi
Regular working pressure : 15 psi

Inner Chamber Dia & Depth	Capacity	Load
250 x 450 mm	22 Liter	2.0 kw
300 x 500 mm	40 Liter	3 kw
350 x 600 mm	50 Liter	3 kw
450 x 600 mm	90 Liter	4 kw

Inner Chamber Dia & Depth	Capacity	Load
550 x 750 mm	187 Liter	5 kw
750 x 1000 mm		6.5 kw

OPTIONAL ACCESSORIES FOR THE ABOVE AUTOCLAVES EXTRA COST

1. Water level indicator, (to indicate water position inside the boiler)
2. Automatic low water level cut of switch
3. Automatic pressure Control cut/off switch
4. Digital temp. Controller-cum-indicator with Sensor
5. Digital timer
5. Radial Locking System

02. BOD Incubator

Specification reference: HSI-104

Technical Specifications

HSI-104 B.O.D. INCUBATOR – SUPER DELUXE TYPE WITH DIGITAL TEMPERATURE CONTROLLER CUM INDICATOR

The BOD incubator is most versatile and highly reliable low temperature incubator to make Biochemical Oxygen Demand determinations and for preservation of vaccines, insulin, level extracts, chemicals etc.

CONSTRUCTION: Double walled robust cabinet, wheel mounted, Insulated with 3" mm super glass wool between the intermediate and exterior walls, which forms the main structure. Outer chamber is made of mild steel attractively finished with powder coating for durable operation. Inner Chamber made of Stainless steel interior walls has supports allowing wide range of shelf positions and spacing. Door fitted with Double glass window and heavy stainless steel hinges facilitate inspection of samples without opening the door. Transparent glass door let viewing the specimen inside without disturbing chamber temperature. Lock arrangements is provided in the double walled outer door. Unit fitted with a door-operated lamp of illumination inside the chamber.

TEMPERATURE CONTROL: B.O.D. Incubator is the excellent and reliable Digital temp. Controller – cum – Indicator with RTD Sensor control range from 5 deg C to 60 deg C +/- 1 deg C. Hermetically sealed high performance Kirloskar-Copeland compressor, air cooled condenser works efficiently to lower the inside chamber temperature and accessories provided at the bottom of the chamber. Heating elements made of Nichrome wires are placed in the inner back side of the BOD Incubator. Cooling coils are distributed all round the chamber and lie in the air circulation path. Air is circulated by a fan to keep the temp uniform throughout inner chamber.

CONTROL PANEL: All controls and circuitry are housed at the top of the incubator and therefore protected from spillage. Separate indicator lamps for mains, heating and cooling are fitted. Temperature setting knob allows the user to select and set any desired temp.

Supplied complete with 2 or 3 shelves of Stainless Steel as per chamber, Cord and plug Suitable to work on 220V, 1 Phaze, 50 Hz, AC supply. The unit is provided with 4 Nos. wheels for easy movements.

Inner Chamber size W x H x D	Capacity Cu. Ft.	Liters	No. of Shelves	Power Rating
455 x 610 x 410 mm	4.0 Cu. Ft.	112	2 Nos.	1.0 KW
505 x 830 x 415 mm	6.0 Cu. Ft.	171	2 Nos.	1.5 KW
565 x 865 x 550 mm	10.0 Cu. Ft.	280	3 Nos.	1.5 KW
650 x 900 x 580 mm	12.0 Cu. Ft.	340	3 Nos.	2.0 KW
700 x 900 x 650 mm	15.0 Cu. Ft.	420	3 Nos.	2.5 KW

OPTIONAL ACCESSORIES

1. Automatic Digital Timer range 0-24 hrs (0-9999)
2. Stabilizer 1 KVA

03. Humidity Chamber

Specification reference: HSI-125

Technical Specifications

HSI-125 "HUMIDITY CHAMBER

Double walled construction, interior made of thick gauge stainless steel sheets and exterior made of cold rolled mild steel sheets finished in powder coat. Inner space in between the walls is packed with special grade glass wool. The door is double walled having toughened glass window and seals against atmospheric infiltration by rubber gasket. The Inner chamber accommodates stainless steel trays at adjustable height.

Humidity is controlled by a humidistat humidity controller, having range 35% to 100% but humidity can be obtained from atmospheric humidity of 95% with sensitivity of ± 3%. Temperature controlled by digital temperature controller from ambient + 5°C to 60°C with an accuracy of ± 1°C.

Heater and blower are fitted in the rear side of the interior for above ambient temperature. For achieving below ambient temperature evaporator coils, compressor, condenser and fan motor are provided. Humidity is attained by the mist of water that is circulated inside the chamber for maintaining 40% to 95% humidity. A water reservoir will be provided with Auto low-level water cut off device and alarm.

Compact control panels with switches, Indicating lamps, Temperature controllers are fixed at the side of the chamber for operational convenience.

Chamber size
455 x 455 x 710 mm
605 x 605 x 605 mm
605 x 605 x 910 mm

04. Incubator - Lab Model

Specification reference: HSI-102

Technical Specifications

HSI-102 BACTERIOLOGICAL INCUBATOR – (LAB MODEL)

Hasthas Incubator reliable day-to-day operation in variety of uses. Drying and staining of slides, paraffin embedding, tissue culture work, incubation of antibody test, excellent for Microbiological determinations etc.

CONSTRUCTION: Bacteriological Incubator is sturdy, double walled with doors, Inner chamber made of Stainless steel. Door fitted with Double glass window and heavy stainless steel hinges facilitate inspection of samples without opening the door. Transparent glass door let viewing the specimen inside without disturbing chamber temperature. The outer body is of thick mild steel sheet with attractively finished in powder coating for durable operation. Inner space in between the walls is filled with 2.5" fine glass wool insulation to minimize the heat loss between the walls.

The temperature is controlled by a capillary thermostat variable from +5 deg above ambient upto 80 deg with a sensitivity of +/- 2 deg C. Inner chamber accommodates easily removable stainless steel perforated trays adjustable height, adjustable Air ventilator is provided at the top.

Supplied complete with trays, air ventilators, pilot lamps indicating Lamps on/off switch, Thermostat, & Cord and plug. Suitable to operate on 220 volts 1 Phase, 50 Hz AC supply.

Available in following chamber sizes:

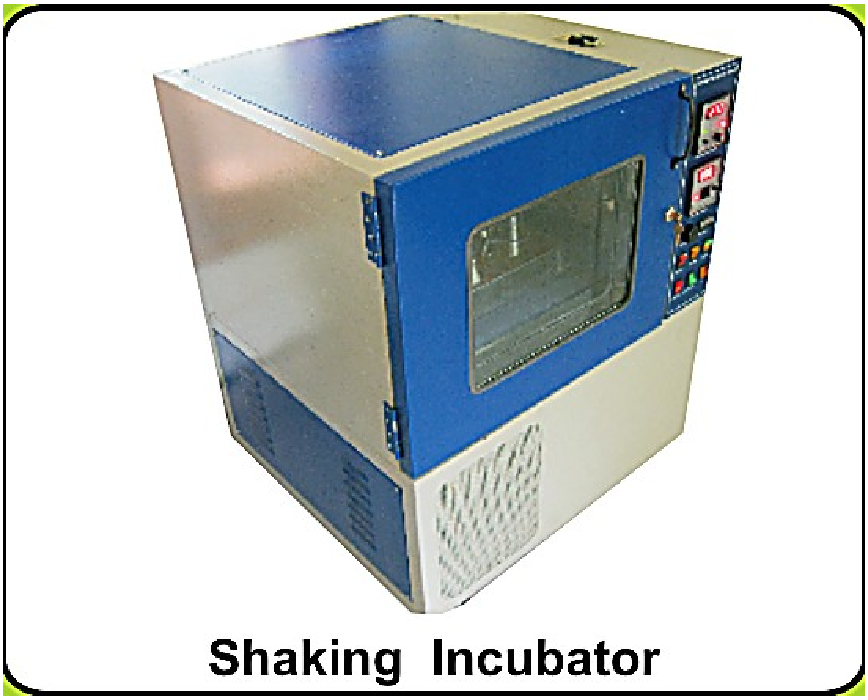
Inner Chamber W x H x D	No. of Shelves	Rating	Liters
350 x 350 x 350 mm	1 No.	300 W	45
450 x 450 x 450 mm	2 Nos.	500 W	95

Inner Chamber W x H x D	No. of Shelves	Rating	Liters
450 x 600 x 450 mm	2 Nos.	600 W	125
600 x 600 x 600 mm	3 Nos.	750 W	224
600 x 900 x 600 mm	3 Nos.	1000 W	336

OPTIONAL ACCESSORIES: SUITABLE FOR THE ABOVE MODEL

1. Digital Temperature Indictor-cum-Controller with RTD Sensor
2. Air Circulating Fan Assembly – (Size: Up to 450 x 450 x 450mm)
3. Air Circulating Fan Assembly – (Size: Up to 450 x 600 x 450mm and Above)
4. Timer range 0-9999 min.

05. Shaking Incubator



Specification reference: HSI-127

Technical Specifications

HSI-127 INCUBATOR SHAKER

Double walled chamber, inner made of polished stainless steel sheets and outer made of cold rolled mild steel sheets with powder coat finish. The wall is insulated with special grade glass wool.

The inner chamber accommodates one no. easily removable stainless steel tray at adjustable height and is also provided with a shaking platform with threaded holes for mounting stainless steel clamps for holding conical flasks from 50 ml to 1000ml capacity.

An outer double door with inner full view acrylic door permits inspection of specimens. Heating is by imported nichrome wires Temp. is controlled by a Digital Indicator cum controller working in conjunction with PT 100 sensor. Powerful air circulator creates positive airflow through the incubator.

The shaker is driven by a variable DC motor with variable speed control. The control panel accommodating, Digital controller, pilot indicating lamps, on/of switches etc, will be provided at the top of the unit.

Specification:

Temperature range	:	5°C to 60°C
Temperature accuracy	:	± 0.5°C
Shaking speed	:	50 to 250 rpm

Platform holding capacity	Platform Size
8 Conical flasks x 250 ml	325mm x 325mm
16Conical flasks x250 ml	425mm x 425mm

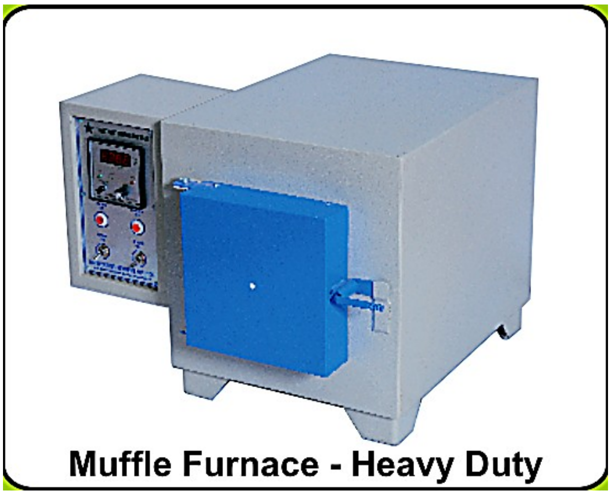
Custom built sizes can also be provided on request.

OPTIONAL ACCESSORIES WILL BE EXTRA

Digital RPM Indicator
Digital Timer (0-9999)
Chamber illumination – two fluorescent lamp
Chamber illumination by two U.V. lamps instead of fluorescent lamp
Voltage stabilizer

06. Muffle Furnace - Standard and Heavy-Duty Models

Standard model



Heavy-duty model

Specification reference: HSI-113

A separate heavy-duty muffle-furnace section was not present in the supplied specification file. HSI-113 is shown as the common/closest matching specification.

Technical Specifications

HSI-113 ELECTRICAL FURNACE GROOVED MUFFLE

Double walled chamber inner made of 4 nos. siliminite refractory slabs and outer made of cold rolled mild steel sheet finished in powder coating. Insulation by Ceramic fibre blankets to minimize radiation heat loss. Double walled insulated door mounted on heavy-duty hinges is provided with effective locking arrangements. Heating elements will be Kanthal A1 coils suspended in the grooves of the refractory.

Temperature is maintained and controlled by Digital temperature controller working in conjunction with Cr/A1 thermocouple placed in the hot zone.Supplied complete with control panel.

Operating Temperature: Ambient +5 to 1100 Deg. C
Max. Temperature : 1150 Deg. C

Light weight with ceramic wool insulation. Suitable to work on 220V, single phase 50 Hz, AC supply.

Muffle Size (inside)	Rating)
100 x 100 x 225 MM (4’’x4’’x9)	1.5 kw
125 x 125 x 250 mm (5"x5"x10")	2.5 kw
150 x 150 x 300 mm (6"x6"x12")	3.5 kw
200 x 200 x 300 mm (8"x8"x12")	5.0 kw
300 x 300 x 300 mm (12'x12'x12")	7.5 kw

07. High Temperature Furnace

Specification reference: HSI-115

Technical Specifications

HSI-115 HIGH TEMPERATURE FURNACE 1400°C

High temperature Muffle Furnaces are designed to meet the requirements of varies customers in Industries, Educational Research fields etc.

Construction: Double walled chamber, outer made of cold rolled mild steel sheets and finished in powder coating paint. Inner chamber formed by high temperature withstand Zirconia Vacuum Board followed by Ceramic fibre blankets on all the sides.

Double walled door mounted on heavy-duty hinges in the front is provided with effective locking arrangements. Heating elements will be Silicon Carbide Rods. Further the door is provided with door limit switch to cut off the power supply whenever the door is opened and to restart when the door is closed.

Temperature is maintained and controlled by Digital PID Temperature controller with Thyristor Control Device working in conjunction with Cr/A1 Thermocouple.Supplied complete with control panel.

Continuous Operating Temperature : 1350 Deg. C
Maximum Temperature : 1400 Deg. C
Suitable to work on 220V, single phase 50 Hz, AC supply.

Muffle size (inside)	Rating
100 x 100 x 225 mm (4'x4"x9")	1.8 kw
125 x 125 x 250 mm (5"x5"x10")	2.5 kw
150 x 150 x 300 mm (6"x6"x12")	3.5 kw
200 x 200 x 300 mm (8"x8"x12")	5.0 kw

Muffle size (inside)	Rating
300 x 300 x 300 mm (12"x12"x12")	7.5 kw
BIGGER SIZES OFFERED ON REQUEST	

Optional: a) Programmable Controller (extra) :

08. Tubular Furnace

Specification reference: HSI-114

Technical Specifications

HSI-114 ELECTRICAL TUBULAR FURNACE 1150°C

The furnace is supplied in vertical or horizontal model, as per request of customers.

Double walled chamber outer chamber made of cold rolled mild steel sheets and finished in powder coating. The Inner chamber is formed by high Alumina tube capable of withstanding temperature up to 1200°C. The tube will have one end closed or both ends opened as per the requirement of customers. Insulation with high-density superior quality ceramic fibre blanket.

The furnace is heated by KANTHAL heating elements. The temperature is controlled by Digital PID Controller with S.S.R. output. The sensor will be Cr/A1 type thermocouple. The size of the tubes available are 40 to 100mm ID and with length of 350mm to 650mm. -

Temperature range: 600 - 1150°C

Power supply: 230 volts, single phase

OPTIONAL ACCESSORIS

- **60-70 ODMM X 600MM Length**

09. Oven - Lab Model

Specification reference: HSI-116

Technical Specifications

HSI-116 HOT AIR OVEN (LABORATORY MODEL)

Construction: It is double walled construction. Inner chamber made of Stainless steel. Outer chamber is made of mild steel attractively finished in powder coating for durable operation. Inner chamber is fabricated with ribs to adjust trays to any convenient height. Oven supplied with removable perforated trays 2 or 3 depending on sizes. Trays are made of Stainless Steel

Insulation: It insulated by 2.5" gap between the walls is filled with special grade glass wool for proper insulation and to avoid heat losses.

Door: It is double walled inner Stainless Steel and outer chamber mild steel with powder coated paint fitted with heavy stainless steel hinges with spring loaded lock.

Heating Elements: Heating elements are made of high-grade Nichrome wire, which are put inside the porcelain beads and placed at the bottom for uniform temperature all over the space.

Temperature Control: The temperature is controlled by a capillary thermostat variable from 5 deg. C above ambient to 250 deg.C with a sensitivity of +/- 2 deg C.

Ventilation: Air ventilators are provided at top as well as at the bottom to ventilate gases and fumes if any.

Control Panel: The equipment is provided with a panel which is just below the door having a Capillary type Thermostat, thermostat control known knob, On/Off switch, two pilot lamps and provision for fixing the Timer.

Power Requirement: Supplied with cord and plug, Suitable to operate on 220V as per indicator single phase, 50 Hz AC supply

Available in following chamber sizes:

Inner Chamber W x H x D	No. of Shelves	Rating	Liters
350 x 350 x 350 mm	1 No.	1.2 KW	45
450 x 450 x 450 mm	2 Nos.	1.8 KW	95
450 x 600 x 450 mm	2 Nos.	2.5 KW	125
600 x 600 x 600 mm	3 Nos.	3.0 KW	224
600 x 900 x 600 mm	3 Nos.	4.0 KW	336

OPTIONAL ACCESSORIES: SUITABLE FOR THE ABOVE MODEL

1. Digital Temperature Indictor-cum-Controller with RTD Sensor
2. Air Circulating Fan Assembly – (Size: Up to 450 x 450 x 450mm)
3. Air Circulating Fan Assembly – (Size: Up to 450 x 600 x 450mm and Above)
3. Timer range 0-9999 min.

10. Oven - Memmert Model

Specification reference: HSI-117

Technical Specifications

HSI-117 HOT AIR OVEN (DELUXE TYPE - MEMMERT MODEL)

Construction: It is **triple** walled construction. Inner chamber made of Stainless steel. Outer chamber is made of mild steel attractively finished in powder coating for durable operation. Inner chamber is fabricated with ribs to adjust trays to any convenient height. Oven supplied with removable perforated trays 2 or 3 depending on sizes. Trays are made of Stainless Steel

Insulation: It insulated by **3"** gap between the walls is filled with special grade glass wool for proper insulation and to avoid heat losses.

Door: It is double walled inner Stainless Steel and outer chamber mild steel with powder coated paint fitted with heavy stainless steel hinges with spring loaded lock.

Heating Elements: Heating elements are made of high-grade Nichrome wire, which are put inside the porcelain beads and placed at the bottom and two vertical sides for uniform temperature all over the space.

Temperature Control: The temperature is controlled by a capillary thermostat variable from 5 deg. C above ambient to 250 deg.C with a sensitivity of +/- 2 deg C.

Ventilation: Air ventilators are provided at top as well as at the bottom to ventilate gases and fumes if any.

Control Panel: The equipment is provided with a panel which is just below the door having a Capillary type Thermostat, thermostat control known knob, On/Off switch, two pilot lamps and provision for fixing the Timer.

Power Requirement: Supplied with cord and plug, Suitable to operate on 220V as per indicator single phase, 50 Hz AC supply.

Available in following chamber sizes:

Inner Chamber W x H x D	No. of Shelves	Rating	Liters
350 x 350 x 350 mm	1 No.	1.2 KW	45
450 x 450 x 450 mm	2 Nos.	1.8 KW	95
450 x 600 x 450 mm	2 Nos.	2.5 KW	125
600 x 600 x 600 mm	3 Nos.	3.0 KW	224
600 x 900 x 600 mm	3 Nos.	4.0 KW	336

OPTIONAL ACCESSORIES: SUITABLE FOR THE ABOVE MODEL

1. Digital Temperature Indictor-cum-Controller with RTD Sensor
2. Air Circulating Fan Assembly – (Size: Up to 450 x 450 x 450mm)
3. Air Circulating Fan Assembly – (Size: Up to 450 x 600 x 450mm and Above)
3. Timer range 0-9999 min.

11. Vacuum Oven

Specification reference: HSI-137

Technical Specifications

HSI-137 VACUUM OVEN

Hasthas Vacuum oven is reliable to-day-to operation in variety of uses in the filed of Medical, Industries etc.

CONSTRUCTION: Unit is study with Double walled chamber. Inner chamber is cylindrical in shape made of Stainless steel with provision for vacuum and the vacuum gauge is provided at the top of unit and outer chamber rectangular in shape made of thick M.S. Sheets attractively finished in powder coating for durable operation.

A glass-viewing window is provided in the centre of the chamber for viewing the specimen inside without disturbing / opening the door. The Oven is filled with superior grade pure white thick glass wool insulation to reduce the radiation heat loss to the minimum. The Inner Chamber accommodates stainless steel trays at adjustable height. The door is of stainless steel plate with specially designed mechanism having alignment. A positive screw specially designed for tightening the lid obtains a perfect seal.

A seal of band heaters is wrapped round the vacuum chamber for fast response and maximum uniformity of heat. The temperature is indicated and controlled by a Digital Temperature Indicator cum controller working in conjunction with a FEK thermocouple.

A compact model control panel-accommodating, pilot indicating lamps, Main Switch, Fuses, Digital Temperature Indicator cum controller working in conjunction with a FEK thermocouple will be fitted on the side of the Oven for easy operation.

Supply 3-core wire and plug. Suitable to operate on 220 volts 1 Phase, 50 Hz AC supply.

Available in following sizes: (SUPPLIED WITHOUT VACUUM PUMP)

Chamber Size	Temperature Range
200 mm Dia x 200 mm Depth	200 Deg.C
250 mm Dia x 250 mm Depth	200 Deg.C
250 mm Dia x 300 mm Depth	200 Deg.C
300 mm Dia x 300 mm Depth	200 Deg.C
300 mm Dia x 450 mm Depth	200 Deg.C
300 mm Dia x 500 mm Depth	200 Deg.C
300 mm Dia x 550 mm Depth	200 Deg.C

12. Industrial Drier

Specification reference: HSI-122

Technical Specifications

HSI-122 INDUSTRIAL DRYING OVEN (TRAY DRIER)

Industrial Tray Driers suitable for heat treatment, baking and drying applications in Industries or Institutes engaged in the production of Vaccines, Tablets, Bottle Sterilizing, Baking of Breads or Biscuits, Drying Chemicals PCB Processing, Electrical or electronic components etc. Application of dry heat with temperature ranging between 50 deg C to 300 deg C.

CONSTRUCTION: Triple walled construction on sturdy angle iron frame with both inner and outer walls of thick PCRC sheet, which is duly decreased, and primer painted to prevent rusting. The inner wall is painted with aluminium paint to withstand long duration heating cycles normally required in Industrial applications.

The 100mm gap between the walls is filled with special grade glass wool for proper insulation and to avoid heat losses thereby saving energy. Air is constantly circulated by heavy-duty blower to maintain the inside temperature constant with a minimum temperature gradient throughout the working chamber.

HEATING ELEMENT: Heating is done by constantly circulating the air with heavy duty blower which are interlocked with Strip type heating elements placed equidistantly on the inner chamber. A ventilator with adjustable opening on the top facilities flowing away of any fumes or vapors produced during the process.

TEMPERATURE CONTROL: A power selection switch which permits selection of working temperature inside the chamber. Temperature from 50 deg C to 300 deg. C +/- 3 deg C is controlled by Automatic Digital temperature controller-cum-Indicator.

VENTILATION: Air ventilators are provided at top as well as the bottom to ventilate gases and fumes if any.

CONTROL PANEL: The equipment is provided with a panel which is having Mains on/off switch 1 No., heater on/off switch 1 no., mains indicating lamps 3 Nos., heater indicating lamps 1 no., motor starter 1 no., (push to on/push to off) ammeters 3 nos., volt meter with selector switch 1 no., heater selector switch 1 no., etc., fitted rights side of the oven itself.

POWER: Suitable to work on 400/440 Volt 3 phase 50 Hz AC mains.

Inner Chamber	Tray Capacity	Power Rating
3' x 3' x 2'	12 Trays	5 kw
3' x 4' x 3'	24 Trays	8 kw
3' x 6' x 3'	48 Trays	12 kw
4' x 8' x 3'	96 Trays	22 kw

OPTIONAL ACCESSORIES

1.	Aluminium Tray size 32"x16"x1½ "
2.	Stainless Steel Tray

13. Deep Freezer - Vertical

Specification reference: HSI-110

Technical Specifications

HSI-110 DEEP FREEZER (VERTICAL)

Hasthas Deep Freezer are reliable day-to-day operation in variety of uses in the field of Medical R&D unit, Agriculture, Industries, Educational Institutions etc.

CONSTRUCTION: Double walled chamber. Inner chamber made of Stainless steel well polished and fusion welded. The outer is made of M.S. Sheets attractively finished in powder coating for durable operation. The inner chamber accommodates Stainless Steel Trays at adjustable height.

INSULATION: High grade glass wool or PUF insulation is provided at the sides between to two walls to minimize the sweating in humid conditions.

Evaporating coils are lead soldered and fixed in the back side of the inner chamber. Double walled door in front provided with magnetic gasket for perfect sealing and provided with heavy duty latching arrangement. Temperature maintained and controlled by Digital Temperature controller working in conjunction with PT 100 sensor. Door operated illumination lamp is provided inside the chamber. Refrigerant system consists of compressor, air cooled condenser and heavy duty motor driven fan provided at the bottom while the control panel with built in digital temp. Controller cum indicator, indicating lamp with switches etc. provided on the top of the cabinet.

TEMPERATURE RANGE : Ambient 5 deg C. to **-20** Deg. C/ **-40** Deg. C

POWER SUPPLY : 250 Volts 1 Phaze, 50 Hz AC Supply

Capacity	Inner chamber Size
110 Litres	425 x 400 x 680 mm
165 Litres	500 x 415 x 825 mm

Capacity	Inner chamber Size
280 Litres	580 x 500 x 1000 mm

Prices for larger models are offered on request

Optional Extra

a) Stabilizer

14. Deep Freezer - Horizontal

Specification reference: HSI-109

Technical Specifications

HSI-109 DEEP FREEZER (HORIZONTAL)

Hasthas Deep Freezer are reliable day-to-day operation in variety of uses in the field of Medical R&D unit, Agriculture, Industries, Educational Institutions etc.

CONSTRUCTION: Double walled chamber with opening at the top. Top portion of the chamber and top lid are made of thick stainless steel well polished and fusion welded. The Outer chamber is made of M.S. Sheet and attractively finished in powder coated finish. Inner chamber is made of Stainless Steel.

INSULATION: High class insulation with combination of special grade glass wool and PUF insulation is provided at the sides between to two walls to minimize the sweating in humid conditions.

The top lid is provided with gasket seals and locking arrangements. Evaporating coils are lead soldered and fixed in the back side of the inner chamber. Temperature maintained and controlled by Digital Temperature controller working in conjunction with PT 100 sensor. Refrigerant system consists of compressor, air cooled condenser and heavy duty motor driven fan are provided at the bottom. The Control panel with built in Digital Temp. Controller cum Indicator, indicating lamp with switches etc. are provided.

TEMPERATURE RANGE	:	Ambient 5 deg C. to **-20 Deg**. C +/- Deg. C
POWER SUPPLY	:	220/240 Volts 1 Phaze, 50 Hz AC Supply

Capacity	Inner chamber Size
110 Litres	550 x 400 x 650 mm
165 Litres	800 x 400 x 650 mm

Capacity	Inner chamber Size
280 Litres	1200 x 400 x 650 mm

Prices for larger models are offered on request

OPTIONAL EXTRA NO DISCOUNT

a) Stabilizer

15. Laminar Flow Cabinets - Vertical and Horizontal

Vertical model

Horizontal model

Specification reference: HSI-128

Technical Specifications

HSI-128 " LAMINAR FLOW CABINETS HORIZONTAL & VERTICAL MODELS

HASTHAS offers a comprehensive range of high performance Horizontal, Vertical Laminar Flow Clean air Cabinets. These are designed to meet the filtration. Illumination, and noise & vibration requirements, providing particle free air to meet class 100 conditions of US Federal Standard.

CONSTRUCTION: All Laminar Flow Cabinets are basically constructed out of DURO BOARD teak wood, which is termite and insect proof, fire and weather resistant. Front, back, top and exterior surfaces are covered with white DECOLAM or FORMICA. Interior are epoxy painted. Work table is made of mica top (optional) S.S. top and side panels of 6mm thick transparent flexi glass duly framed.

AIR FLOW AND FILTRATION: Laminar flow principle involves double filtration of air. Atmospheric air is drawn through pre filter and is made to pass through highly effective HEPA (High Efficiency Particular Air) filters having efficiency rating as high as 99.99% with cold DOP and 99.97% with hot DOP, thus retaining all air-borne particles of size 0.3 micron and larger. Double filtered air blow in laminar flow through the work table at designed velocity of 90 ft/min +/- 20%.

BLOWER MOTOR ASSEMBLY: Statically and dynamically balanced, direct drive, sized to provide adequate air flow volume over the entire surface of HEPA filter. Fitted with 1/4 HP motor and operates with minimum noise level i.e. lower than 65 db on scale and Vibration less than 2.5 um. (0.0001 deg).

LIGHTING: Work area properly illuminated by diffused, glare free fluorescent light supplied with Gas inlet, Front Door, Manometer, Wheel for easy movement, UV Lamp.

POWER REQUIREMENT: 230V, Single phase, 50 Hz AC supply

Size: Horizontal & Vertical

Model	Model	HL-22/VL-22	HL-32/VL-32	HL-42/VL-42
Working Area		2' x 2' x 2'	3' x 2' x 2'	4' x 2' x 2'
Size of HEPA Filter		2' x 2' x 6"	3 x 2' x 6"	4 x 2' x 6"
No.of HEPA Filter		1	1	1
No.of Pre Filter		1	2	2
Illumination		1x20w	1x20w	2x40w

16. Fume Hood

Specification reference: HSI-111

Technical Specifications

HSI-111 FUME EXHAUST HOOD

Laboratory fumes hoods are one of the most important components used to protect laboratory personnel from exposure to hazardous chemicals and agents used in the laboratory. When used properly, they exhaust toxic, offensive or flammable materials when the containment of an experiments or procedures fails and vapors or dust escape from the apparatus being used.

Constructions: Cabinets is made of Thick Ply Board / M.S. Duty Powder coated / Stainless steel (304). Interiors coated with Asbestos / Lead sheets and further coated with fire proof epoxy paint. Working table upto 2 feet is covered with glazed ceramic Acid Resistant tiles. Working table is covered with Granite / Tiles / S.S. 304.

Door: Front door is made of 6mm clear Toughened glass. Counter weight balance Front Door. Vertical sliding facility & the front door can be stopped at any point.

Exhaust System: Exhaust blower consists of single phase motor which is enclosed in a wooden casing connected with a PVC Exhaust Ducting.

Working Area Size
2' x 2' x 2'
3’ x 2’ x 2’
4' x 2' x 2'
5’ x 2’ x 2’
6' x 2' x 2'

17. Water Bath - Thermostatic and Digital Control

Thermostatic model

Digital-control model

Specification reference: HSI-139, HSI-140

The supplied specifications list 6-hole and 12-hole laboratory water baths. Digital temperature control is listed as an optional accessory.

Technical Specifications

HSI-139 WATER BATH – LAB TYPE (6 Holes Rectangular)

Double walled chamber, Inner SS, Outer M.S. / S.S. with powder coat finish, immersion heater, thermostat control, Concentric rings, Glass wool insulation.

Temperature Range: 5 Deg C ambient to 90 Deg. C.
Size: 350 x 250 x 100mm (6 Holes)

Chamber Size	Watts
Double Walled with Thermostat (Outer M.S)	1000
Double Walled with Thermostat (Full S.S)	1000
OPTIONAL: **Digital Temp. Controller-cum-Indicator at extra cost**	**each**

HSI-140 WATER BATH – LAB TYPE (12 Holes Rectangular)

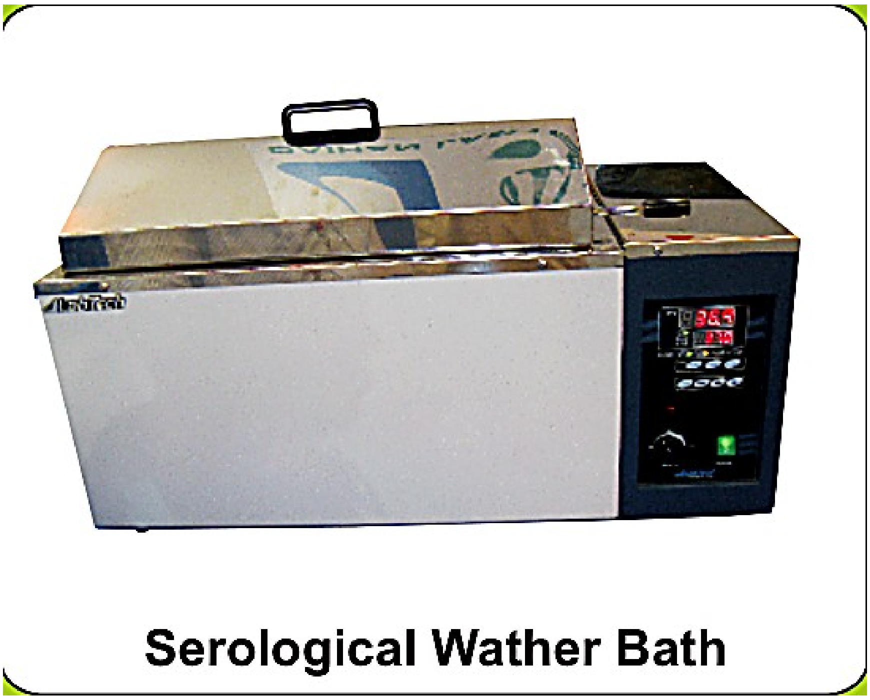
Double walled chamber, Inner SS, Outer M.S. / S.S. with powder coat finish, immersion heater, thermostat control, Concentric rings, Glass wool insulation.

Temperature Range: 5 Deg C ambient to 90 Deg. C.
Size: 400 x 300 x 100mm (12 Holes)

Chamber Size	Watts
Double Walled with Thermostat (Outer M.S)	1500
Double Walled with Thermostat (Full S.S)	1500

Chamber Size	Watts
<u>OPTIONAL:</u> **Digital Temp. Controller-cum-Indicator at extra cost**	each

18. Serological Water Bath



Specification reference: HSI-141

Technical Specifications

HSI-141 SEROLOGICAL WATER BATH

HASTHAS Serological Water baths are versatile enough to handle any clinical procedure, Incubation, Inactivation, Agglutination, as well as most serological pharmaceutical, biomedical procedure.

CONSTRUCTION: Double wall construction. Complete inner chamber made of HIGHLY POLISHED STAINLESS STEEL. Outer Chamber is made of Mild Steel Sheet, finished with power coating. Gap between the walls is filled with special grade glass wool for proper insulation to avoid heat losses.

The Water Bath is provided with a drain plug to facilitate easy emptying and cleaning of the inner chamber whenever necessary. Pyramidal shaped cover and perforated removable diffuser are standard accessories.

HEATING ELEMENTS: Immersion heating elements made of high grade materials are fitted at bottom with different rating for different sizes.

TEMPERATURE CONTROL: Temperature is generally controlled by Capillary thermostat from 5 deg. ambient to 90 deg.C +/- 1 deg.C.

CONTROL PANEL: The equipment is provided with a panel having a thermostat. ON/OFF switch, two pilot lamps. Supplied with cord and plug.

POWER: Suitable to operate on 220V, Single Phase, 50 Hz, AC supply.

Chamber Size	Load
250 x 175 x 175 mm for 2 racks	1 KW
350 x 250 x 175 mm for 4 racks	1.5 KW
450 x 250 x 175 mm for 6 racks	2 KW
600 x 300 x 175 mm for 8 racks	3 KW

OPTIONAL ACCESSORIES

1. Stirrer with 1/20 hp Motor with S.S. rod and blade
2. Digital temperature Controller

19. Constant Temperature Bath

Specification reference: HSI-107

Technical Specifications

HSI-107 CONSTANT TEMPERATURE BATH

Double walled chamber, inner made of thick stainless steel sheets and outer made of mild steel sheets duly finished in powder coating. Glass Window provided in the front side for an easy inside view.

Inner space in between the walls tightly packed with special grade glass wool. Top Lid is made of stainless steel sheet plate and provided with handle. Heating elements are chromium plated immersion type. Temperature controlled by precision thermostat. A medium speed stirrer with 1/20 HP motor is fitted to the unit for continuous stirring and thus maintaining uniform temperature throughout the working chamber. Built in control panel accommodates indicating lamps, precision thermostat, On/Off switch, speed regulator to control speed of the stirrer. Designed to work on 220/230 volts a/c supply.

TEMPERATURE RANGE: Ambient +5 deg C to 100 deg C +/- 2 deg C

Capacity Volume	Chamber size
20 Litres	200 x 250 x 320 mm
24 Litres	220 x 250 x 320 mm
40 Litres	370 x 260 x 350 mm

20. Low Temperature Water Bath

Specification reference: HSI-129

Technical Specifications

HSI-129 LOW TEMPERATURE BATH

Hasthas Low temperature Baths are reliable day-to-day operation in variety of uses in the field of Medical, R & D units, Processing Industries etc.

CONSTRUCTION: Unit is sturdy with Double walled chamber. Inner chamber made of stainless steel and Outer chamber made of cold rolled M.S. Sheets attractively finished in powder coating for durable operation.

Insulation	:	70 mm P.U.F. Insulation between two walls.
Heating element	:	'U' type tubular immersion heater
Cooling	:	Refrigeration system attached
Circulation	:	Circulation pump for uniform temperature
Control	:	Digital proportional temperature controller with P1 100 Sensor.
Power Supply	:	240 volts 1 Phaze, 50 Hz AC supply

Temperature Range
-10°C to 90°C
-20°C to 90°C
-30 °C to 90°C

Higher size c apacity prize on request.

0

21. Hot Plate - Circular

Hot Plate - Circular

Specification reference: HSI-123

Technical Specifications

HSI-123 HOT PLATE (CIRCULAR)

This is heavy duly heating plate with a removable MS/SS plate at the top. The housing is made of mild steel sheet and finished in powder coat paint. A heat control switch for low, medium and high heat and two nos. indicating lamps are provided on the front panel

The hot plate has a maximum power consumption of 3.0 Kw. The unit is designed for operating on 230 V, 1 phase, 50 Hz AC supply.

Standard size	Rating
8" Dia	1.5 kw
10" Dia	2.0 kw
12" Dia	2.0 kw

22. Hot Plate - Rectangular

Specification reference: HSI-124

Technical Specifications

HSI-124 HOT PLATE (RECTANGULAR)

Standard Size	Rating
10" x 12" Dia	1.5 kw
10" x 16" Dia	1.5 kw
12" x 18" Dia	1.5 kw
12" x 24" Dia	2.0 kw
18" x 24" Dia	3.5 kw

23. C.O.D Heating Block

Specification reference: HSI-120

The supplied file contains HSI-120 "Soxhlet Extraction Mantle (C.O.D.)" rather than a separately titled C.O.D. heating-block section. It is included as the closest name match and should be technically confirmed before final printing.

Technical Specifications

HSI-120 SOXHLET EXTRACTION MANTLE (C.O.D) (WITHOUT GLASS PARTS)

Capacity	Watts	Type
250 ml	450	3T x 3R
250 ml	900	6T x 6R
500 ml	600	3T x 3R
500 ml	1200	6T x 6R

24. Magnetic Stirrer

Specification reference: HSI-130

The supplied specification for HSI-130 is brief and identifies the model as a magnetic stirrer with hot plate.

Technical Specifications

HSI-130 MAGNETIC STIRRER

TYPE
With Hot Plate

25. Flocculator

Specification reference: HSI-112

Technical Specifications

HSI-112 FLOCCULATOR (JAR TESTING APPARATUS)

Illuminator base consists of fluorescent tube mounted below translucent plastic to provide diffused cold light through floc samples. Flocculator consists of geared continuous run heady duty 1/20 H.P. variable speed motor from 50 to 100 RPM with built in speed control.

Stainless steel stirring rods are provided with adjustable spacers to adjust the depth of stirring paddles. The stirring shaft can be removed without disturbing other stirrers. It is supplied with stirrers (2, 4, or 6 depending upon order).Suitable to operate on 220 Volt, 50Hz, 1 Phase, AC supply. The units are supplied with beakers.

SIZE
TWO STIRRER
FOUR STIRRER
SIX STIRRER

OPTIONAL ACCESSORIES

- DIGITAL RPM METER

26. Heating Mantle

Specification reference: HSI-119

Technical Specifications

HSI-119 HEATING MANTLE WITH REGULATOR:

Aluminium Housing with powder coat finish. The heating elements are made out of Kanthal wire on helically wound coil form safety insulated by fibre glass sleevings which in turn is attached to knitted glass fabric, to give flexible support to the flask and controlled by a simmerstat.

Temperature range: 5 deg.C above ambient to 400 deg. C

Capacity	Watts
100 ml	60
250 ml	150
500 ml	200
1 Litres	300
2 Litres	500
3 Liters	500
5 Litres	800
10 Litres	1000
20 Litres	1500

27. Bunsen Burner

Specification reference: HSI-105

Technical Specifications

HSI-105 BUNSEN BURNER

Electrically operated, Body made of Stainless Steel and the base made of Stainless sheet. Unit is fitted with built-in Energy Regulator and indicating lamps. Supplied with plug and cord. Available with/without Pyrometer. Suitable for heating crucibles with samples.

Temp. Range: 600 deg. C

Type of Power Rating
350 Watts (With regulator)

28. Water Distillation Units

Wall-mounting water still

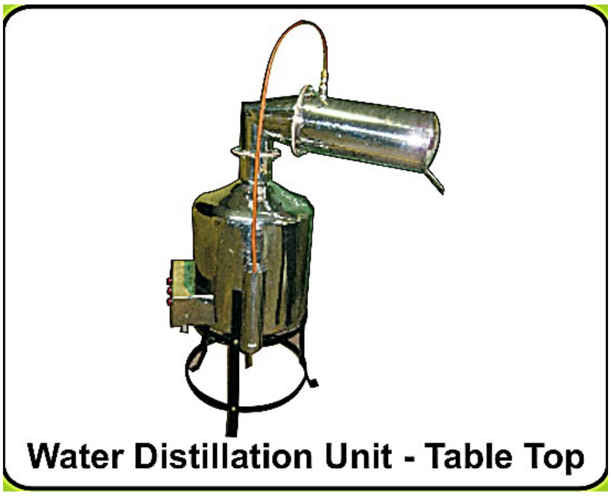


Table-top water distillation unit

Specification reference: HSI-143

The supplied HSI-143 specification describes a single wall-mounting water distillation unit. A separate table-top specification was not present.

Technical Specifications

HSI-143 WATER DISTILLATION UNIT – Single

Wall mounting type. The entire fabrication is made of Stainless steel and supplied in highly polished finish. The unit comprises of a main boiling chamber with a high dome lid and provided with water sealing arrangement and a tubular condensing column with an automatic water level arrangement to maintain the water level inside the chamber always constant.

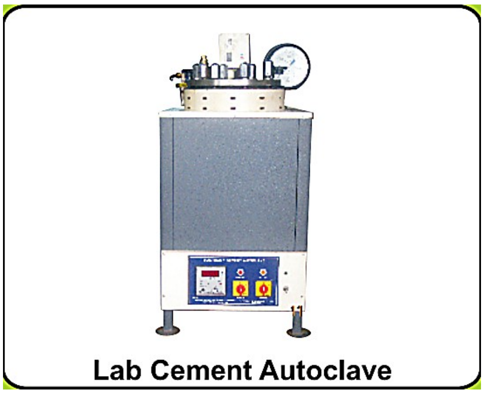
Capacity (Ltrs/hr)	Power in KW
2	2
4	3
6	4.5
8	6
10	8

Appendix: Products Requiring Separate Specifications

The following products are shown in the supplied brochure, but no clearly matching specification section was found in the supplied technical specification document. Their images are included here so the missing specifications can be added later.

Autoclave Horizontal

Matching technical specification not found in the supplied file.



Lab Cement Autoclave

Matching technical specification not found in the supplied file.

CO2 Incubator

Matching technical specification not found in the supplied file.

Hot and Cold Chamber

Matching technical specification not found in the supplied file.

Bio-safety Cabinet

Matching technical specification not found in the supplied file.

Viscosity Bath

Matching technical specification not found in the supplied file.

Curing Tank

Matching technical specification not found in the supplied file.

GENERAL TERMS AND CONDITIONS

01	Price	: Ex-Godown, Chennai-32
02	Sales tax	: GST will be charged as applicable as per rule
03	Delivery	: 2 to 4 weeks from the date of receipt your P.O.
04	Packing & Forwarding	: Will be extra
05	Payment	: 50% advance along with your purchase order 50% against delivery
06	Freight charges	: To – Pay basis
07	Installation	: For heavy instruments & require personal Installation and demonstration at site by our engineer, will be charged extra based on mutual Discussion.
08.	Guarantee	: One year from date of deliver against my manufacturing defects only.

SPECIAL INSTRUCTION FOR DEALERS

Please provide us the consignee address, their TIN and C.S.T. Nos, & Your TIN and C.S.T. Nos. If the dispatch is to be made to places other than your city. In the absence of same we shall not be responsible for any check post detention.